

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 02



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Recap of Previous Lecture



Topic

Propositions and their types



Topic

Connectives



Topic

Tautology, Contradiction, Contingency and Satisfiable propositional functions



Topics to be Covered



★
Topic

Logical Implication

★
Topic

Logical Equivalences

★
Topic

Argument / Inference

★
Topic

Rules of inference

★
Topic

Practice questions



Topic : Logical Implication / Implication

Let P and Q are any two propositional functions.

Whenever P is true if Q is also true then P logically implies Q .

P logically implies Q if and only if $P \rightarrow Q$ is a tautology.

* Check whether P logically implies Q is Valid or invalid

Ans • Whenever P is true if Q is also true, then

P logically implies Q is valid.

* If there exists at least one case such that P is true but Q is false then P logically implies Q is invalid



Topic : Logical Equivalence / Equivalence

- If P and Q are any two propositional functions, then P equivalent to Q is written as $P \equiv Q$. $\textcircled{\text{or}}$ $P \simeq Q$ $\textcircled{\text{or}}$ $P \leftrightarrow Q$
- P and Q are said to be equivalent if and only if they have same truth table.
- $P \equiv Q$ if and only if $P \leftrightarrow Q$ is a tautology.
- $P \equiv Q$ if and only if P logically implies Q and Q logically implies P
- $P \leftrightarrow Q$ is a tautology if and only if, $P \rightarrow Q$ is a tautology as well as $Q \rightarrow P$ is a tautology



Topic : Some important equivalences

① $\rightarrow \sim(\sim P) \equiv P$ } Double Complementation

② $\rightarrow \left. \begin{array}{l} P \vee Q \equiv Q \vee P \\ P \wedge Q \equiv Q \wedge P \end{array} \right\}$ Commutative

③ $\rightarrow \left. \begin{array}{l} (P \vee Q) \vee R \equiv P \vee (Q \vee R) \\ (P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R) \end{array} \right\}$ Associative



Topic : Some important equivalences

$$\textcircled{4} \quad P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

} Distributive Property

$$\textcircled{5} \quad \sim(P \wedge Q) \equiv (\sim P \vee \sim Q)$$

$$\sim(P \vee Q) \equiv (\sim P \wedge \sim Q)$$

} De' Morgan law

$$\textcircled{6} \quad P \wedge (P \vee Q) \equiv P$$

$$P \vee (P \wedge Q) \equiv P$$

} Absorption law



Topic : Some important equivalences

$$\begin{array}{l} \textcircled{7} \quad P \wedge P \equiv P \\ \quad \quad P \vee P \equiv P \end{array} \quad \left. \vphantom{\begin{array}{l} P \wedge P \equiv P \\ P \vee P \equiv P \end{array}} \right\} \text{Idempotent law}$$

$$\begin{array}{l} P \vee T \equiv T \\ P \wedge F \equiv F \end{array} \quad \left. \vphantom{\begin{array}{l} P \vee T \equiv T \\ P \wedge F \equiv F \end{array}} \right\} \text{Domination law}$$

$$\begin{array}{l} P \vee F \equiv P \\ P \wedge T \equiv P \end{array} \quad \left. \vphantom{\begin{array}{l} P \vee F \equiv P \\ P \wedge T \equiv P \end{array}} \right\} \text{Identity}$$

$$T \wedge T \equiv T$$

$$T \vee T \equiv T$$

$$T \wedge F \equiv F$$

$$T \vee F \equiv T$$

$$F \vee F \equiv F$$

$$F \wedge F \equiv F$$

+ Some important Equivalences :-

➤ $P \rightarrow Q \equiv \sim P \vee Q$

• $P \rightarrow Q \equiv \sim Q \rightarrow \sim P$

• $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$



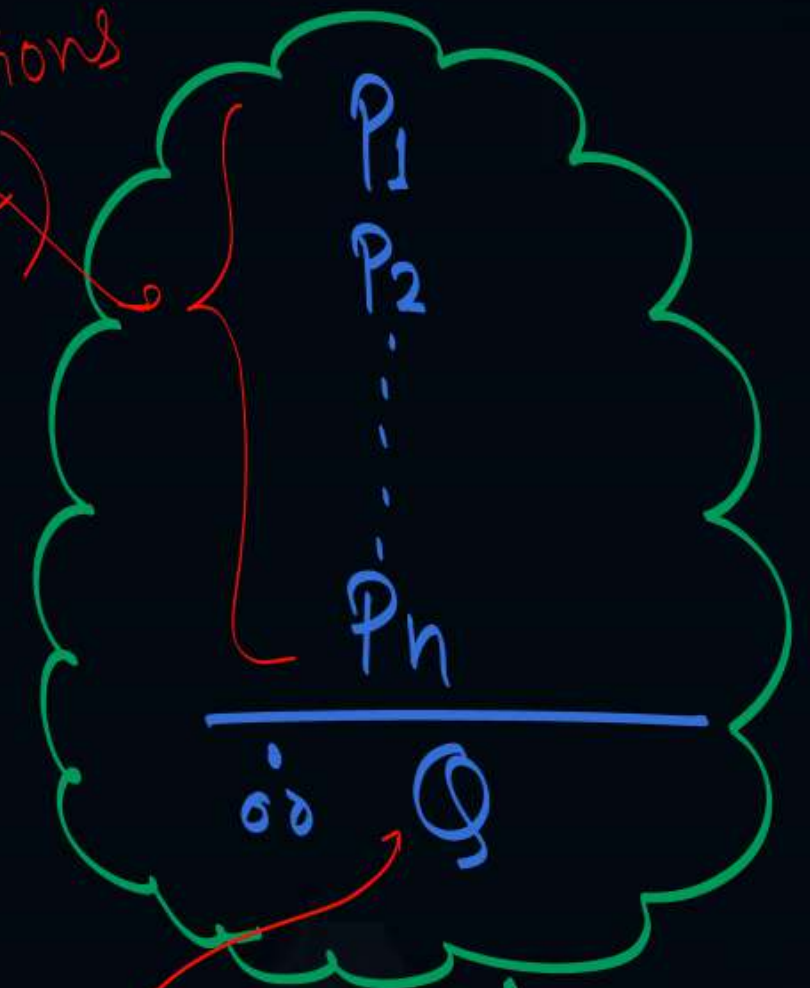
Topic : Argument / Inference



- ✦ The statement that,
A set of premises $\{ P_1, P_2, P_3, \dots, P_n \}$
yields another proposition Q is called
an argument.

The argument is, that whenever Propositions $P_1, P_2, \dots, P_n$
are true then we can conclude that proposition
 Q will also be true.
 Q is called conclusion of the argument.

Propositions
(Premise)



Conclusion

it is a way
to represent

an argument

• Argument may be valid
or invalid.

These are the given statements
They are always considered to be true

P_1 : If it rains today then Cricket match will not be played

P_2 : Cricket match is not played

Q : It rained today

Conclusion

it is an argument

given argument may be valid
or may be invalid.

p : it rains today
 q : Cricket match will not be played

$p \rightarrow q$

q

$\neg p$

it is an invalid argument

if $p \rightarrow q$ is true
and q is also true,
then p may be true or
 p may be false



Topic : Argument / Inference



- ④ Argument may be valid or invalid.
- ④ The process of reasoning whether the argument is valid or invalid is called inference.
- ④ If conclusion Q can be inferred from the set of premises by applying some rules of inference and equivalences, then argument is said to be valid otherwise invalid.



Topic : Argument / Inference

Following statements are equivalent, yields

- Argument $\{P_1, P_2, P_3, \dots, P_n\} \vdash Q$ is valid.
 $\equiv$ Conclusion Q follows from the premises $\{P_1, P_2, P_3, \dots, P_n\}$
- $\{P_1, P_2, P_3, \dots, P_n\}$ Logically implies Q is valid
 $\{P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n\}$ logically implies Q is valid
- $\{P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n\} \rightarrow Q$ is a tautology

If we can prove the validity of any one out of these statements, then other two will follow.



Topic : Rules of inference



Any valid *reasoning* is rule of inference.



Topic : Rules of inference



1. Simplification

$$\frac{p \wedge q}{\therefore p} \text{ is valid}$$

Similarly, $\frac{p \wedge q}{\therefore q} \text{ is valid}$

i.e, $(p \wedge q) \vdash p$ is valid

i.e, $(p \wedge q)$ logically implies p is Valid

i.e $(p \wedge q) \rightarrow p$ is a tautology.



Topic : Rules of inference



2. Addition

$$\frac{p}{\therefore p \vee q} \text{ is valid}$$

Addition can only be performed using disjunction

$$\frac{p}{\therefore p \wedge q} \text{ is invalid}$$

$$\frac{p \vee q}{\therefore p} \text{ is invalid}$$

Simplification can only be performed w.r.t. conjunction



Topic : Rules of inference



3. Conjunction

$$\frac{p \quad q}{\therefore p \wedge q} \text{ is valid}$$



Topic : Rules of inference



4. Disjunctive Syllogism

$$\begin{array}{c} p \vee q \\ \sim p \\ \hline \therefore q \end{array} \text{ is valid}$$

$$\begin{array}{c} p \vee q \\ \sim q \\ \hline \therefore p \end{array} \text{ is valid}$$



Topic : Rules of inference



5. Conjunctive Syllogism

$$\sim(p \wedge q) \quad \equiv$$
$$\frac{p}{\therefore \sim q} \text{ is valid}$$

$$\sim p \vee \sim q$$
$$\frac{p}{\therefore \sim q}$$



Topic : Rules of inference



6. Modus Ponens

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline q \end{array} \text{ is valid}$$

7. Fallacy of affirming the consequent

$$\begin{array}{c} p \rightarrow q \\ q \\ \hline p \end{array} \text{ is invalid}$$

{ it is not necessary for 'p' to be true }



Topic : Rules of inference



8. Modus Tollen's

$$\begin{array}{c} p \rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array} \text{ is valid}$$

9. Fallacy of denying the antecedent

$$\begin{array}{c} p \rightarrow q \\ \sim p \\ \hline \therefore \sim q \end{array} \text{ is invalid}$$



Topic : Rules of inference



10. Transitivity

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r} \text{ is valid}$$



Topic : Rules of inference



11. Dilemma

$$\begin{array}{l} p \rightarrow r \\ q \rightarrow r \\ p \vee q \\ \hline \therefore r \end{array} \text{ is valid}$$



Topic : Rules of inference



12. Constructive Dilemma

$$p \longrightarrow r$$

$$q \longrightarrow r$$

$$p \vee q$$

$$\therefore r \vee r$$

is valid



Topic : Rules of inference

13. Destructive Dilemma

$$p \rightarrow r$$

$$q \rightarrow s$$

$$\sim r \vee \sim s$$

is valid

$$\therefore \sim p \vee \sim q$$

$\sim r$	$\sim s$	r	s	p	q	$\sim p$	$\sim q$
T	T	T	T	T	T	F	F
T	T	T	T	F	T	T	F
T	T	T	F	T	F	F	T
T	T	T	T	T	F	F	T
T	T	F	T	T	T	F	F
T	T	F	F	T	F	F	T
T	F	T	T	T	T	F	F
T	F	T	F	T	F	F	T
T	F	F	T	F	T	T	F
T	F	F	F	F	T	T	F
F	T	T	T	T	T	F	F
F	T	T	F	T	F	F	T
F	T	F	T	T	T	F	F
F	T	F	F	T	F	F	T
F	F	T	T	T	T	F	F
F	F	T	F	T	F	F	T
F	F	F	T	F	T	T	F
F	F	F	F	F	T	T	F

At least one of
 $\sim p$ (or) $\sim q$
will be true



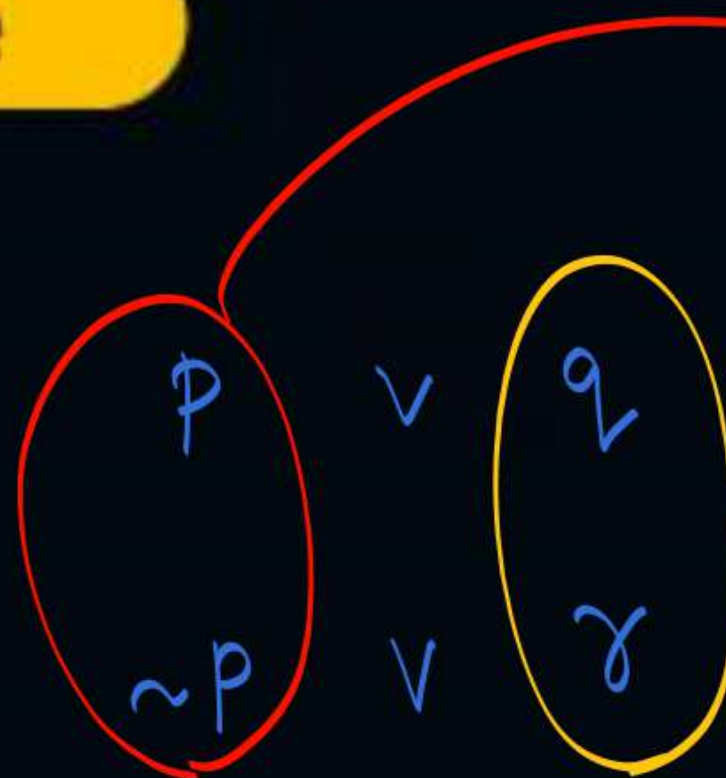
Topic : Rules of inference

14. Resolution

$$p \vee q$$

$$\sim p \vee r$$

$$\hline \therefore q \vee r \text{ is valid}$$



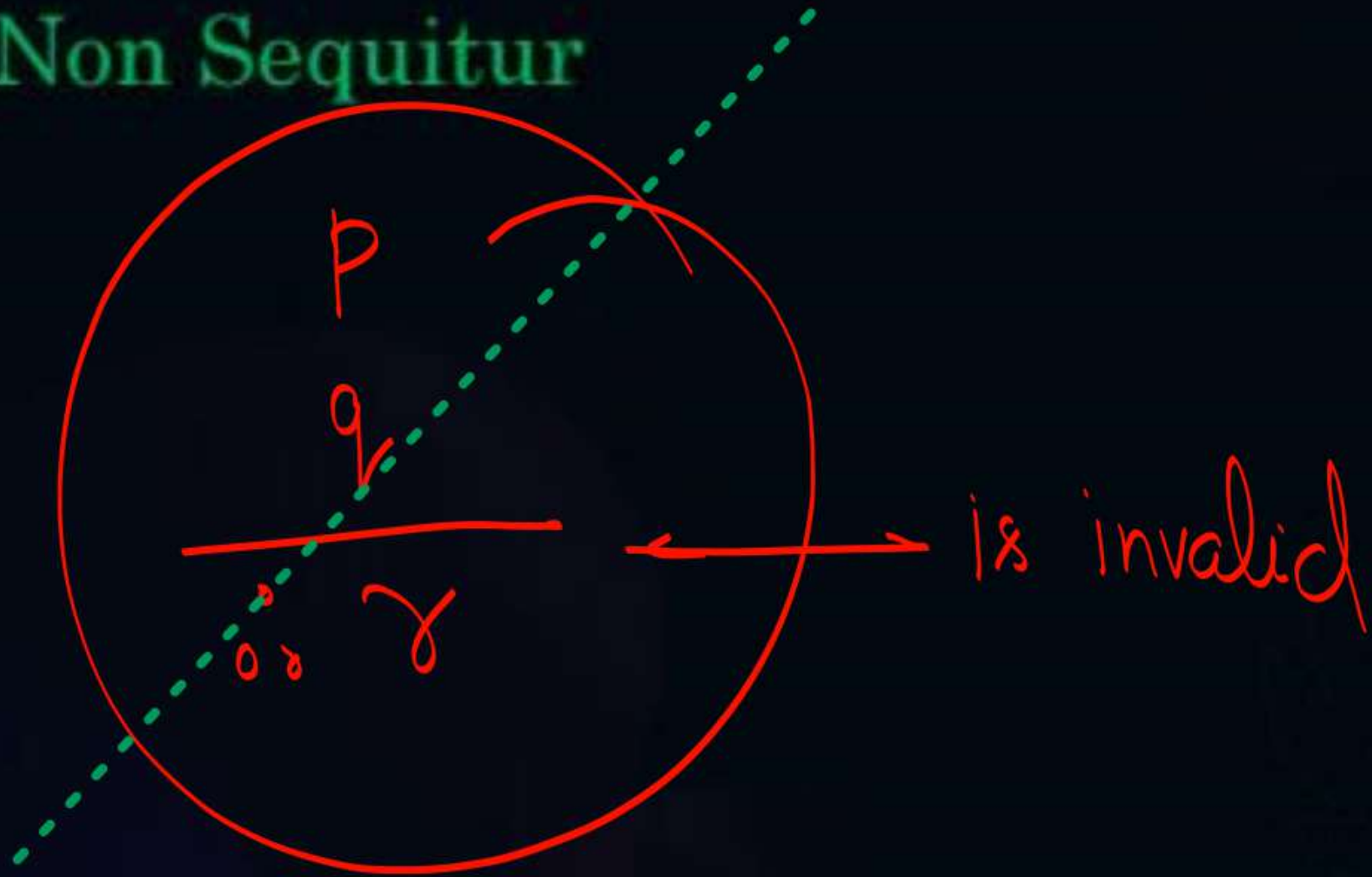
at least one of p or $\sim p$ will be false

∴ At least one of q or r must be true w.r.t. false value of p or $\sim p$



Topic : Rules of inference

15. Non Sequitur



Q. Simplify the following Statement

$$\{\sim p \wedge (\sim q \wedge r)\} \vee \{(q \wedge r) \vee (p \wedge r)\}$$

$$\overset{|||}{\{(\sim p \wedge \sim q) \wedge r\}} \vee \overset{|||}{\{(q \vee p) \wedge r\}}$$

$$\underbrace{\{\sim(p \vee q) \wedge r\}}_x \vee \underbrace{\{(p \vee q) \wedge r\}}_y$$

$$(x \wedge r) \vee (y \wedge r)$$

$$(x \vee y) \wedge r = \underbrace{\{\sim(p \vee q) \vee (p \vee q)\}}_T \wedge r$$

$$T \wedge r = r$$

(a) True

(b) False

(c) p

(d) q

☒ (e) r

✓ Q: Simplify the following Expression

$$\left[(P \vee Q) \wedge \underbrace{\sim \{ \sim P \wedge (\sim Q \vee \sim R) \}} \right] \vee \left[(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R) \right]$$

$$\left[(P \vee Q) \wedge \{ P \vee (Q \wedge R) \} \right] \vee \left[\sim (P \vee Q) \vee \sim (P \vee R) \right]$$

$$\left[(P \vee Q) \wedge \{ (P \vee Q) \wedge (P \vee R) \} \right] \vee \sim \left[(P \vee Q) \wedge (P \vee R) \right]$$

$$\left[\{ (P \vee Q) \wedge (P \vee Q) \} \wedge (P \vee R) \right] \vee \sim \left[(P \vee Q) \wedge (P \vee R) \right]$$

$$\underbrace{\left[(P \vee Q) \wedge (P \vee R) \right]}_x \vee \sim \underbrace{\left[(P \vee Q) \wedge (P \vee R) \right]}_x$$

$\equiv \text{True}$

H.W. Q: Which of the following is/are valid?

- a) $(P \wedge Q)$ logically implies $(P \leftrightarrow Q)$
- b) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow Q)$
- c) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow \sim Q)$
- d) Q logically implies $(P \rightarrow Q)$

H.W. Q: Which of the following is/are tautology?

a) $(P \wedge Q) \longrightarrow (P \longleftrightarrow Q)$

b) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow Q)$

c) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow \sim Q)$

d) $Q \longrightarrow (P \longrightarrow Q)$

H.W. Q: Which of the following is/are tautology?

(a) $(a \vee b) \rightarrow (b \wedge c)$

(b) $(a \wedge b) \rightarrow (b \vee c)$

(c) $(a \vee b) \rightarrow (b \rightarrow c)$

(d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$

H.W. Q. Consider the following statements

$$S_1: \{(A \wedge B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$$

$$S_2: \{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$$

then which of the following is true ?

- (a) Both S_1 & S_2 are valid
- (b) Only S_1 is valid
- (c) Only S_2 is valid
- (d) Neither S_1 nor S_2 is valid.

H.W. Q. Which of the following is are true?

(a) $\{a \rightarrow (b \vee c)\} \equiv \{(a \wedge \sim b) \rightarrow c\}$

(b) $\{(P \rightarrow Q) \wedge (R \rightarrow Q)\} \equiv \{(P \vee R) \rightarrow Q\}$

(c) $\{(P \rightarrow Q) \vee (R \rightarrow Q)\} \equiv \{(P \wedge R) \rightarrow Q\}$

(d) All of the above

H.W. Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above propositional statement is.

- (a) Satisfiable but not valid
- (b) Valid
- (c) Contradiction



2 mins Summary



Topic

Logical Implication

Topic

Logical Equivalences

Topic

Argument / Inference

Topic

Rules of inference

Topic

Practice questions

THANK - YOU